

Pre-Calculus Concepts



Limits: an introductory numerical approach

- 1 Estimate $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$ by first simplifying the expression algebraically.
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Average rate of change

- 2 Find the average rate of change of $f(x) = x^2 + 1$ over the interval $[1, 4]$.
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Conic sections: ellipses

- 3 Find the equation of an ellipse centered at the origin with $a = 5$ (horizontal semi-axis) and $b = 3$ (vertical semi-axis).
- 4 For the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$, find the coordinates of the vertices along the major axis.
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Conic sections: hyperbolas

- 5 Find the equation of a hyperbola centered at the origin with vertices at $(\pm 4, 0)$ and $b = 3$.
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Parametric equations

- 6 Eliminate the parameter from $x = t + 1$ and $y = t^2$, expressing y in terms of x .
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Sequences and series: sigma notation

- 7 Evaluate $\sum_{k=1}^5 (2k + 1)$.
- 8 Write $4 + 8 + 12 + 16 + 20$ using sigma notation, and evaluate it.
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Binomial theorem

- 9 Use the binomial theorem to expand $(x + 2)^4$.

- 10 Find the coefficient of x^3 in the expansion of $(x + 3)^5$, using $\binom{5}{k}$.
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Vectors in the plane

- 11 Given $\mathbf{u} = (3, -4)$, find its magnitude and a unit vector in the same direction.
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A multi-part conic and function analysis problem

- 12 This question has two parts. A circle has equation $x^2 + y^2 - 4x + 6y - 12 = 0$. (a) Complete the square to write the equation in standard form, and state the center and radius. (b) Determine whether the point $(6, -1)$ lies inside, on, or outside the circle.
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Matrix operations

- 13 Matrix A has rows $(2, -1)$ and $(3, 4)$; matrix B has rows $(0, 5)$ and $(-2, 1)$. Find $A - B$.
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Infinite geometric series

- 14 Find the sum of the infinite geometric series $8 + 4 + 2 + 1 + \dots$, using $S = \frac{a_1}{1-r}$.
- 15 Write the repeating decimal $0.\overline{6}$ as a fraction, by treating it as an infinite geometric series $\frac{6}{10} + \frac{6}{100} + \frac{6}{1000} + \dots$.
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Piecewise and step functions in applied contexts

- 16 A shipping company charges 5 dollars per pound for the first 10 pounds, then 3 dollars per pound for each pound beyond 10. Write a piecewise function $C(w)$ for the cost, and find the cost of a 15-pound package.
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Finding terms of a sequence defined recursively

- 17 A sequence is defined by $a_1 = 3$ and $a_n = 2a_{n-1} + 1$ for $n \geq 2$. Find a_4 .
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A multi-part conic section identification and analysis problem

- 18** This question has three parts. An equation is given: $4x^2 + 9y^2 - 16x + 18y - 11 = 0$. (a) Complete the square in both x and y to write the equation in standard conic form. (b) Identify the type of conic section. (c) State the center and the lengths of the semi-axes.